

Clinical Note Summarization Without Hallucination

Submitted to Dr.Dinesh Naik.
Amit Nishad-262IT002
Nikhilesh Joshi -262IT029
NITK, Mtech-IT

Introduction — Clinical Notes & Hallucination

What are Clinical Notes?

- **Digital records written by doctors** after examining patients
- **Contains:**
  - Patient history,
  - Symptoms,
  - Diagnosis,
  - Medicines,
  - Lab reports,
  - Treatment plan

What is Hallucination?

- **AI generates information** not present in the original note
- **Intrinsic:** changes existing facts (e.g., age 30 → 45)
- **Extrinsic:** adds new unverified facts (e.g., heart surgery)

Goal: Reduce documentation burden while preserving medical factuality.

Example —

Clinical note: 40-year-old male with fever, cough, diabetes history. Prescribed Paracetamol and antibiotics.

Generated summary: Patient has fever and cough, with diabetes history. Treatment includes Paracetamol and antibiotics.

Motivation

Problems with the Current System

- Hundreds of notes daily
- Very lengthy & repetitive
- Time-consuming
- Hard to understand quickly

AI Can Help, But…

- Current AI may add false facts
- Misses medicines / changes diagnosis
- Produces unsafe summaries

Our Motivation

Build a trustworthy summarization system — **Faithful AI + Reliable Summary**



Better Patient Safety
Fewer factual errors in records



Faster Clinical Decision Support
Quick, reliable summaries



Trustworthy Medical AI
Faithful to the source note

Literature Review — Papers Studied

Research Paper	Contribution
Clinical Safety Framework(2025)	Checks clinical summaries for **hallucinations and missing information**.
MEDAL(ACL 2024)	**Finds and corrects hallucinations** in medical summaries
LLM Clinical Note Summarization System (2025)	Uses **NER + LLM** to find important clinical information and create summaries.

Learnings: LLM summaries can hallucinate; hallucination detection methods exist; better data improves faithfulness; and correct basing with strong LLMs is essential.

Literature Outcome

Finding	Gap
Good summaries generated	Not always factually correct
Powerful LLMs available	Can invent medical facts
High ROUGE scores	ROUGE doesn't measure hallucination
Fluent output	Human evaluation still necessary

Conclusion: Literature reveals a gap between linguistic quality and factual correctness.

Problem Statement

Current Issue — AI systems summarize clinical notes but…

- Add false diseases
- Miss important medicines
- Change diagnosis
- Produce unsafe summaries

Problem Statement

"Existing clinical summarization models generate fluent summaries but cannot guarantee factual faithfulness, making hallucination a major challenge in healthcare AI."

Key Risks & Needs

Hallucination Risk

False diagnoses, medicines, or patient history

Patient Safety

Incorrect summaries can affect treatment decisions

Need for Faithfulness

Summarize only what exists in the original note

Objectives

1

Generate concise summaries

Short but complete

2

Reduce hallucination

No false information

3

Preserve medical facts

Diagnosis, medicine, symptoms correct

AI Pipeline



Focus: Summarization quality + Factual reliability

Methodology — Overview



1. Dataset

MIMIC-III / MIMIC-IV

- Patient history
- Diagnosis
- Medication
- ICU records

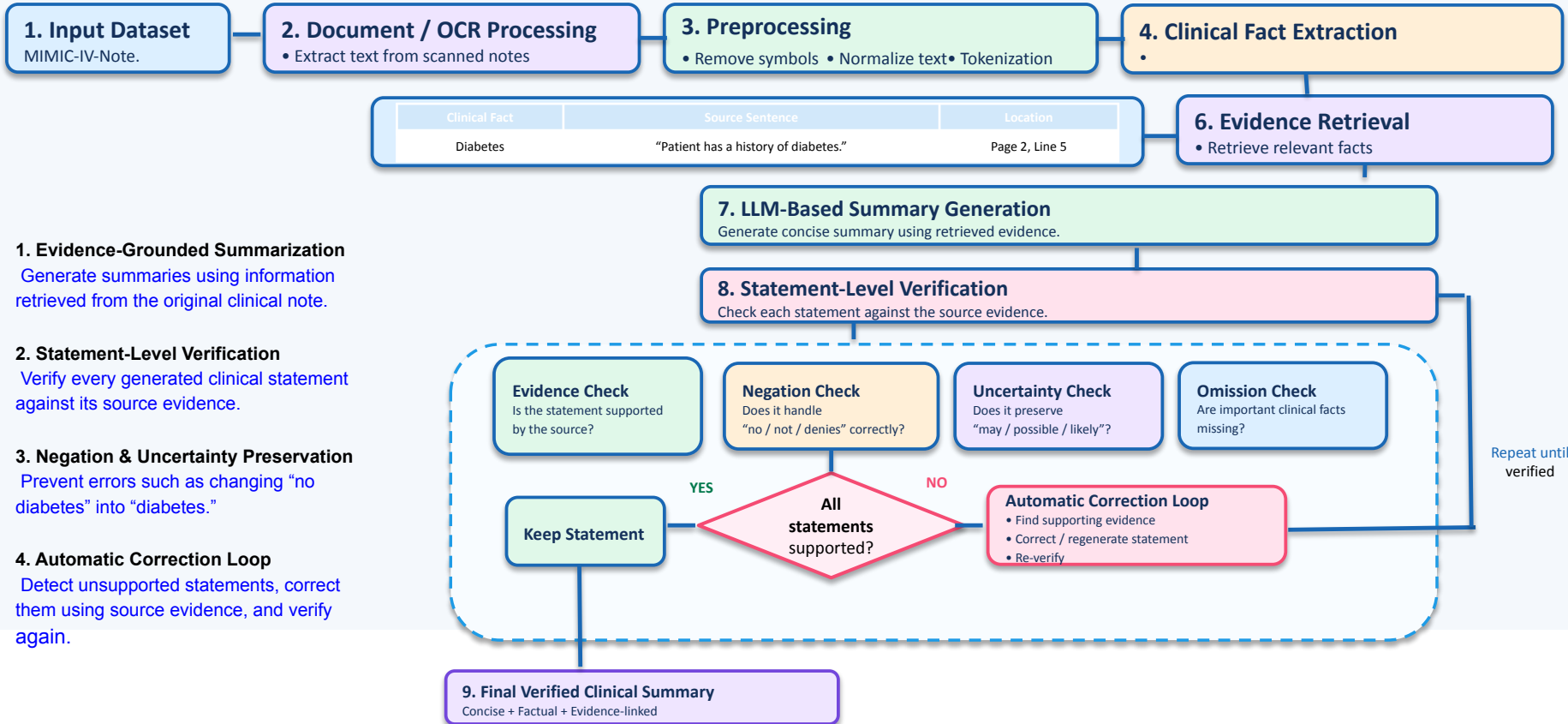
2. Preprocessing

- Remove symbols
- Normalize text
- Tokenization
- Sentence segmentation
- OCR for scanned docs

3. CV + NLP

- **CV Tasks:**
  OCR, layout detection, enhancement, text region
- **NLP Tasks:**
  entity extraction, context understanding

Proposed Novelty



1. Evidence-Grounded Summarization
Generate summaries using information retrieved from the original clinical note.

2. Statement-Level Verification
Verify every generated clinical statement against its source evidence.

3. Negation & Uncertainty Preservation
Prevent errors such as changing "no diabetes" into "diabetes."

4. Automatic Correction Loop
Detect unsupported statements, correct them using source evidence, and verify again.

Thank You